HINTS AND SOLUTIONS

Q.1 C

Q.2 B

Sol. $ev_1 = hc\left[\frac{1}{\lambda} - \frac{1}{\lambda_o}\right]$ . . . (1)

$ev_2 = hc\left[\frac{1}{2\lambda} - \frac{1}{\lambda_o}\right]$ . . . (2)

From equation (1) and (2)

$$3 = \left[\left(\frac{1}{\lambda} - \frac{1}{\lambda_o}\right)\Big/\left(\left[\frac{1}{2\lambda} - \frac{1}{\lambda_o}\right]\right)\right] \Rightarrow \frac{3}{2\lambda} - \frac{1}{\lambda} = \frac{3}{\lambda_o} - \frac{1}{\lambda_o} = \frac{2}{\lambda_o}$$

or, $\frac{1}{2\lambda} = \frac{2}{\lambda_o} \Rightarrow \lambda_0 = 4\lambda$

Q.3 D

Q.4 B

Sol. $T \propto n^3$

Q.5 D

Q.6 B

Q.7 A

Q.8 C

Q.9 A

Q.10 B

Q.11 D

Q.12 A

Sol. $L_1 > L_0$

$\Rightarrow E_{t_1} < E_{t_0} \quad \Rightarrow \quad \lambda_{t_1} > \lambda_{t_0}$

$\Rightarrow$ red light can be absorbed and infrared scattered.

Q.13 A

Sol. $620 = \frac{1240}{E_t}$

$E_t = 2eV = \frac{h^2}{8mL^2}$

$L^2 = \frac{h \times h}{8m}$

$E_t = \frac{1240 \times 6.6 \times 10^{-34}}{2 \times 9.1 \times 10^{-34} \times 2} = 2.27 \times 10^{-10}$ m

Q.14 D

Sol. λ_{min} n= $\dfrac{1242}{V}$

Q.15 Ans. (A)-P (B)-P (C)-R]

Q.16 $E = (13.6\text{ eV})\ Z^2\left(\dfrac{1}{(Z+2)^2} - \dfrac{1}{Z^2}\right)$

$$= -13.6 \times \left(\frac{Z^2-(Z+2)^2}{(Z+2)^2}\right) = \frac{4(Z+1)\times 13.6}{(Z+2)^2}\text{eV} \qquad ...(i)$$

Now energy of electron is $k = \dfrac{h^2}{2\lambda^2 m}$; $\left(\textit{we have } \lambda = \dfrac{h}{\sqrt{2mk}}\right)$

Or, $k = 6\text{eV}$

So, $\dfrac{4(Z+1)\times 13.6}{(Z+2)^2} = 6 + 4.2 = 10.2\text{ eV}$

Or, $\dfrac{(Z+1)}{(Z+2)^2} = \dfrac{3}{16} \Rightarrow (Z-2)\ (3Z+2) = 0$

So, the value of Z = 2 (neglecting the negative/ fractional value)

Q.17 Ans. 30940 Å, 1289 Å, 7000 S

Sol. From Wein's law,

$\lambda_A T$ = const.

$\therefore\ \lambda_A \times 1500 = \lambda_A \times 3600 \qquad ...(i)$

$K_A = \dfrac{hc}{\lambda_a} - 1.2 \qquad ...(ii)$

$K_B = \dfrac{hc}{\lambda_b} - 1.2 \qquad ...(iii)$

$\therefore\ \dfrac{K_A}{K_B} = \left(\dfrac{(hc/\lambda_a)-1.2}{(hc/\lambda_b)-1.2}\right) = 1/3$

$\dfrac{\lambda_a}{3\lambda_b} = \dfrac{12375 - 1.2\lambda_a}{12375 - 1.2\lambda_b}$

$4 \times 12375 - 4.8 l_b = 5 \times 12375 - 6\ \lambda_a$

$6\ \lambda_a - 4.8\lambda_b\ = 12375$

$\left(\dfrac{12\times 6}{5} - 4.8\right)\lambda_b\ = 12375$

$9.6\ \lambda_b\ = 12375$

$\lambda_b\ = \dfrac{12375}{9.6}A^\circ\ = 1289A^o$

$\lambda_a\ = 3093.6\ A^o$

$N = \dfrac{10^{12}}{10^6}\ = 10^6$/sec. (no. of electrons emitted per sec.)

$V = 9.6 - 1.2 = 8.4$ Volts

$\therefore \dfrac{Q}{4\pi\varepsilon_0 r} = 8.4$

$$\frac{10^6 \times 9\times10^9 \times 1.6\times10^{-19} \times t}{12\times10^{-2}} = 8.4$$

$\therefore$ $t = 700$ sec.

Q.18 Ans. 15.7 eV, 7 km/s

Sol. Momentum of the photon ,$p = \dfrac{E}{c} = \dfrac{(1.33\times10^6)(1.6\times10^{-19})}{3\times10^8}$

$= 7.09 \times 10^{-22}$ kgms^{-1}

Using momentum conservation recoil energy of the

^{60}Ni is $E = \dfrac{p^2}{2m} = \dfrac{(7.09\times10^{-22})^2}{2(60\times1.67\times10^{-27})} = 2.51\times 10^{-18}$ J $= 1.57 \times 10^{-5}$ MeV

Recoil speed of the nucleus, $v = \dfrac{p}{m} = \dfrac{7.09\times10^{-22}}{60\times1.67\times10^{-27}} = 7.07\times10^3\,m/s$

Q.19 Ans. 2.52×10^{19} eV, $3.3 \times 10–8$ N

Sol. (i) Energy of each photon $= E = \dfrac{hc}{\lambda} = 3.975 \times 10^{-19}$ J

No. of photons falling on surface per second & being absorbed,

$n = \dfrac{10J}{2.48eV} = 2.52 \times 1019$ eV

(ii) The linear momentum of each photon $= p = \dfrac{h}{\lambda} = \dfrac{h\nu}{c}$

$\therefore$ Total momentum of all photons (falling in one sec.)

$= \dfrac{nh\nu}{c} = \dfrac{10J}{3\times10^8} = 3.33 \times 10^{-8}$ N-s

Rate of change of momentum $=$ Force $= \dfrac{dp}{dt} = 3.33 \times 10^{-8}$ N.

Q.20 [Ans. (a) 0.367 curie (b) 0.633 curie]